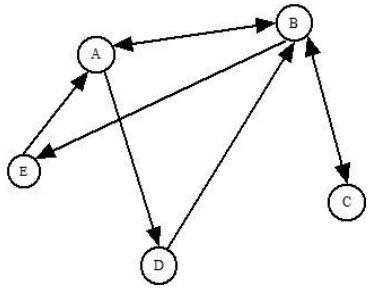
American Computer Science League

2020-2021 ● Contest 4: Solutions ● Classroom Division

1. Graph Theory

The directed graph is as follows:



The paths of length 2 can be found by inspection: BAB, BCB, BAD, BEA.
The number of paths can also be found by squaring the adjacency matrix:

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Adding the numbers in the second row gives 4 paths from B of length 2.

1. 4 (D)

2. Graph Theory

The cycles from A are: ABA, ACA, ABCA, ABDA, ABDCA.

2. 5 (C)

3. Graph Theory

Cubing the adjacency matrix will give the number of paths of length 3.

$$\begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}^3 \end{array} = \begin{bmatrix} 3 & 0 & 3 & 3 \\ 3 & 1 & 3 & 1 \\ 3 & 0 & 2 & 3 \\ 3 & 0 & 3 & 2 \end{bmatrix}$$

A, C and D do not have paths of length 3 to vertex B.

3. A, C, D (A)

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4. Graph Theory

There are 8 cycles in the directed graph: ABA, AEA, ABCA. ADBA, ADBCA, AEDBA, AEDBCA, and BDB. Remember that BAB is not a new cycle.

4. 8 (D)

5. Digital Electronics

The digital circuit can be represented by the Boolean expression:
$A + (AB + B)$
$= A + B(A + 1)$
$= A + B$

5. $A + B$ (B)

6. Digital Electronics

The digital circuit can be represented by the Boolean expression:

$$(\overline{(A + B)(BC)})\,\overline{C} = (\overline{A} + \overline{B} + \overline{BC})\,\overline{C}$$
$$= (\overline{A}\,\overline{B} + \overline{B} + \overline{C})\,\overline{C}$$
$$= \overline{A}\,\overline{B}\,\overline{C} + \overline{B}\,\overline{C} + \overline{C}\,\overline{C}$$
$$= \overline{C}\,(\overline{A}\,\overline{B} + \overline{B} + 1)$$
$$= \overline{C}$$

To be TRUE, $\overline{C}$= 1, so $C = 0$, $A = *$, $B = *$.
There are 4 ordered pairs: (1, 1, 0), (1, 0, 0), (0, 1, 0), (0, 0, 0).

6. (*, *, 0) (A)

7. **Digital Electronics**

The Boolean expression represented by the digital circuit is:

$$(\overline{A + (\overline{AB})(B + C)})\,C = (\overline{A}\,(\overline{(\overline{AB})(B + C)})\,C$$
$$= \overline{A}\,C\,((\overline{\overline{AB}}) + (\overline{B + C}))$$
$$= \overline{A}\,C\,(AB + \overline{B}\,\overline{C})$$
$$= A\,\overline{A}\,B\,C + \overline{A}\,\overline{B}\,C\,\overline{C}$$
$$= 0 + 0 = 0$$

Since the expression is always FALSE, none, or 0, make it TRUE.

7. 0 (A)

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8. Digital Electronics

The Boolean expression represented by the digital circuit is:

$(A \oplus B)(B \oplus C) = (\overline{A}B + A\overline{B})(\overline{B}C + B\overline{C})$

$= \overline{A}\,B\,\overline{B}\,C + \overline{A}\,B\,B\,\overline{C} + A\,\overline{B}\,\overline{B}\,C + A\,\overline{B}\,B\,\overline{C}$

$= 0 + \overline{A}\,B\,\overline{C} + A\,\overline{B}\,C + 0$

$= \overline{A}\,B\,\overline{C} + A\,\overline{B}\,C$

Either $\overline{A}\,B\,\overline{C} = 1 \Rightarrow A = 0,\ B = 1,\ C = 0 \Rightarrow (0, 1, 0)$

or $A\,\overline{B}\,C = 1 \Rightarrow A = 1,\ B = 0,\ C = 1 \Rightarrow (1, 0, 1)$

8. (0, 1, 0) (1, 0, 1) (E)

9. Assembly Language

This is a translation of the assembly language program:

```
n = 47
p = n / 10 (integer division)    p = 4
s = (p * 10 - n) * -1 + p        s = (40 - 47) * -1 + 4 = 11
output s                         This prints the sum of the 2 digits.
```

9. 11 (C)

10. Assembly Language

This is a translation of the assembly language program:

```
x = 21 : y = 49
while y !=  0
        r = x - int(x / y) * y
        x = y
        y = r
end while
output x
```

This is the Euclidean Algorithm to find the greatest common divisor of two numbers. The GCF of 21 and 49 is 7.

10. 7 (B)